

MPCS Lab Bench Test Simulation Master Manual

System: Medha Servo Drives (MDS 736 / MCS 657) - WAG7/WAP4 Electric Locomotive System

1. Historical & Technical Summary

Evolution from Relay-Based Control: Conventional electric locomotives traditionally relied on intricate networks of electromechanical relays (such as Q44, Q118, Q60, etc.) and massive copper wiring harnesses to manage vital functions like circuit breaker (DJ) closing, tap changer progression, and safety monitoring. These legacy systems suffered from high diagnostic latency, frequent mechanical failure points, and lacked any data logging capabilities.

Introduction of MPCS: The Microprocessor-Based Control and Fault Diagnostic System (MPCS) modernized locomotive operations by replacing hardware relay chains with robust digital and software logic loops. Monitored inputs are sampled as 110V DC signals via specialized Input Cards, processed by the central processor within milliseconds, and directed to Output Cards to fire contactor coils. Crucially, systems like the Medha MDS 736 introduced non-volatile fault memory tracking, allowing maintenance staff to extract precise historical fault logs with time stamps.

2. Bidirectional Master Time-Sequence Matrix

Execute the following digital inputs precisely according to the time windows to avoid CPU Time-out Errors (Er) or safety trips.

Step / Time	Action / Inputs to Change	Forward Mode	Reverse Mode	System Output & Response
Step 01 0 Sec (Boot)	Initial Health Setup: Turn ON base safety switches.	I80=1 (Loco Sel 1) I49-I54=1 (Safety Relays) I04-I14=1 (Iso Switches) I20=1 (Range Guard - Lock ON) I19=1 (GR-0) I25=1 (ASMGR Notch) I16=0 (BN), I35=0 (DJ_FB)	Same as Forward Mode	MPCS display boots up. Shows Notch 00 and 'Ready for DJ Close' status. No active fault logs.
Step 02 +2 Sec	VCD Permanent Bypass: Block the vigilance penalty timer.	I126 = 1 (VCD_RESET) Lock this switch to HIGH for the duration of the test.	Same as Forward Mode	VCD internal timer freezes permanently. Prevents sudden buzzer alarms or safety penalty brake triggers.
Step 03 +5 Sec	Pneumatic Setup: Simulate sufficient air pressure.	I65 = 1 (QPDJ_FB) I48 = 1 (BL Key)	Same as Forward Mode	Simulates 6.5 kg/cm² pneumatic operating pressure. Microprocessor permits circuit breaker operations.
Step 04 +6 Sec	DJ Closing Command: Trigger the breaker impulse.	I0 = 1 (BLDJ ON) Toggle I1 (BLRDJ) = 1 for exactly 1 second, then return to 0 (LOW) .	Same as Forward Mode	CPU fires Digital Output Card 1: 00/O4 (DJ Output / MTDJ Coil) turns HIGH (Live LED turns ON).
Step 05 +7 Sec (Critical)	DJ Feedback Loop: Establish breaker state. <i>*Must execute within 1.0s of Step 04!</i>	Toggle I35 (DJ_FB) = 1 (HIGH)	Toggle I35 (DJ_FB) = 1 (HIGH)	Red alarm clears from display. Status safely locks into ' DJ CLOSED ' mode. (Delay >1.5s trips system).

Step / Time	Action / Inputs to Change	Forward Mode	Reverse Mode	System Output & Response
Step 06 +12 Sec	Converter Alignment: Simulate starting phase completion.	Toggle I63 (C118 N/C) = 0 (LOW) <i>*Execute exactly 5 seconds after Step 05.</i>	Toggle I63 (C118 N/C) = 0 (LOW)	Simulates ARNO starting contactor opening after reaching full speed. Output O-75 (WORKING) turns solid green.
Step 07 +15 Sec	Direction Locking: Establish cab running orientation.	Set I27 (MPJ_FOR) = 1 Keep I32 (REV) = 0	Set I32 (MPJ_REV) = 1 Keep I27 (FOR) = 0	Direction locks properly. O10 (J1-J2 UP) fires for Forward; O11 (J1-J2 DN) fires for Reverse. No Mismatch Fault.
Step 08 +18 Sec	Progression Command: Initiate notch advancement.	Toggle I34 = 1 [MP (+)]	Toggle I34 = 1 [MP (+)]	Output Card 1 triggers O14 (VE-UP Coil) to command physical tap changer progression.
Step 09 +19 Sec (Dynamic)	Notch Transition Phase: Simulate mechanical air motor movement.	1. Turn I25 (ASMGR Notch) = 0 2. Pulse I16 [ASMGR(BN)] = 1 for 1 sec, then return to 0 3. Turn I19 (GR-0) = 0	Same as Forward Mode	CPU registers that the tap changer is traveling between physical notch boundaries.
Step 10 +20 Sec	Locking Notch 01: Complete sequence. <i>*Steps 8-10 must finish within 3-5 seconds.</i>	Lock I25 (ASMGR Notch) = 1 (HIGH)	Lock I25 (ASMGR Notch) = 1 (HIGH)	Display Unit safely locks into Notch 01 without 'Er' code. Line contactors (O26/O27) and blowers (O5, O7, O6) fire sequentially.
Reset	Return to Neutral: Safely reset bench to zero.	Set I34 = 0 Set I19 = 1 Set I27 = 0	Set I34 = 0 Set I19 = 1 Set I32 = 0	Display returns to Notch 00 safely. Output O30 (LSGR Green LED) illuminates to confirm absolute neutral.

3. Vital Diagnostic Software Logic Rules

- **I20 (Range Guard) Absolute Requirement:** As defined on page 23 of the software architecture logs, digital input I20 (ASMGR IL 61/62, Ferrule 077) must remain HIGH continuously. Leaving I20 at 0 flags an out-of-bounds mechanical limit error, completely inhibiting the O0/O4 DJ closing loop.
- **I25 (ASMGR ON Notch) Dynamic Transition:** Unlike static bypass lines, I25 (Ferrule 082) cannot be permanently tied high. It must drop to 0 whenever a progression command is given, pulse I16 (BN), and return to 1 upon landing on the target notch. A stagnant I25 signal tells the processor the tap changer failed to move, tripping a notch progression timeout error ('Er').
- **Time-Critical I35 Window:** The processor expects the DJ feedback input I35 to flip from 0 to 1 within a tight 1.0-second window following the O0/O4 output impulse. Failure to toggle I35 within this microsecond window initiates an automated trip sequence to protect the circuit.
- **Directional Complementarity:** The forward selection line I27 (Card 2, Pin N) and reverse selection line I32 (Card 3, Pin A) share an exclusive-OR relationship. Activating both simultaneously generates a Direction Mismatch Fault, immediately venting control air via the emergency coil outputs.
- **Footnote Health/Medical Disclaimer:** This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.